

A DISTRIBUTION-FREE TWO-SAMPLE TEST FOR DISPERSION FOR SYMMETRICAL DISTRIBUTIONS

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1. Summary

The test of Mood (1954) is a well-known test for the equality of two sample variances. When the two sample sizes are different this test is in general also consistent against differences only in location. While this test is designed for the case where the underlying distributions do not differ in location, the property mentioned above is regarded as undesirable, the reason being that in practice the assumption of equality of means is not always realistic. It has been suggested that Mood's test may be applied to pseudo-observations obtained by reducing each observation by the sample median, and Crouse (1964) showed that this modified test is asymptotically distribution-free. However, one may expect distributional difficulties in the case of small samples. This raises the question of whether Mood's test can be modified to overcome consistency against differences in location, and in this paper it is shown that this can be achieved for symmetrical underlying distributions.

The proposed modified test possesses the same a.r.e. properties as Mood's test.

Critical values of the modified test are presented, and a small Monte Carlo study is reported.

2. The consistency properties of Mood's test

Let $X_1, \dots, X_m; Y_1, \dots, Y_n$ be two independent random samples from distributions with continuous distribution functions F and G respectively. Let $r_1, \dots, r_m$ be the ranks of the X 's in the combined sample, then Mood's test is based on

$$M = \sum_{i=1}^m \left(r_i - \frac{m+n+1}{2} \right)^2$$

with large and/or small values of M determining the critical region.

Let p_{YXY} denote the probability of an X -observation falling between two Y -observations, i.e.

$$p_{YXY} = 2P \{ Y_1 < X_1 < Y_2 \},$$

and similarly

$$p_{XYX} = 2P \{ X_1 < Y_1 < X_2 \},$$

then in general (see section 3)

$$(1) \quad E(M) - E(M | F \equiv G) = \frac{1}{2} mn \{ (m-1)(p_{XYX} - 1/3) - (n-1)(p_{YXY} - 1/3) \}.$$

Furthermore (cf. Mood (1954))

$$\text{var}(M|F \equiv G) = \frac{1}{180} mn(m+n+1)(m+n+2)(m+n-2).$$

Theorem 1

If, as $(m+n)$ tends to infinity, $\lim_{m+n} \frac{m}{m+n}$ exists and is bounded away from one and zero, and Z_α is the upper α -point of the standard normal distribution, then the test with critical region

$$\frac{M - E(M|F \equiv G)}{\sqrt{\text{var}(M|F \equiv G)}} > Z_\alpha$$

is consistent if and only if

$$\lim_n \frac{1}{n} \left\{ (m-1) \left(p_{XYX} - \frac{1}{3} \right) - (n-1) \left(p_{YXY} - \frac{1}{3} \right) \right\} > 0.$$

The proof of Theorem 1 follows along the same lines as the proof of Theorem 4 of Section 4. Clearly a similar result holds for the two-sided test.

Consider

$$G(x) = F(x+k),$$

i.e. F and G differ only in location, then generally

$$p_{XYX} \neq \frac{1}{3}, p_{YXY} \neq \frac{1}{3},$$

and hence, if $\lim_n \frac{m}{n} \neq 1$, the test based on large values of M can be consistent against such alternatives. Clearly similar results hold for two-sided tests.

As an illustration consider $m = 2$, $n = 9$, with the hypothesis to be tested $\sigma^2_X = \sigma^2_Y$ against $\sigma^2_X > \sigma^2_Y$, at 10% significance level. From Laubscher et al (1968) the following sequences of X 's and Y 's form the critical region:

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X Y Y Y Y Y Y Y Y X
X Y Y Y Y Y Y Y X Y
Y X Y Y Y Y Y Y Y X
X X Y Y Y Y Y Y Y Y
Y Y Y Y Y Y Y Y X X.
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Obviously the last two sequences do not necessarily reflect a difference in dispersion but more probably a difference in location, and hence if extreme differences in location are present one will often be rejecting the hypothesis even if it is true.

Apart from this unwanted property one would also wish (1) to be of the form

$$h_1(mn)h_2(p_{XYX}, p_{YXY}),$$

because then the class of consistent alternatives is only determined by the population dispersion parameters p_{XYX} and p_{YXY} .

3. A modified form of Mood's test

Let $M_{XYX}(M_{YXY})$ denote the number of times that a $Y(X)$ appears be-

tween two X's (Y's) in the combined sequence of X's and Y's. The statistic M_{XYX} can be written as

$$M_{XYX} = \sum_{i_1=1}^m \sum_{i_2=1}^m \sum_{j=1}^n t(X_{i_1}, X_{i_2}; Y_j)$$

where
$$t(a, b; c) = \begin{cases} 1 & \text{if } c \text{ lies between } a \text{ and } b \\ 0 & \text{otherwise.} \end{cases}$$

Furthermore $E(M_{XYX}) = \binom{m}{2} n p_{XYX}$ and hence $\binom{m}{2}^{-1} n^{-1} M_{XYX}$ is an unbiased estimator of the dispersion parameter p_{XYX} . Similarly $\binom{n}{2}^{-1} m^{-1} M_{YXY}$ is an unbiased estimator of the complementary dispersion parameter p_{YXY} .

Theorem 2

$$M = M_{XYX} + M_{YXY} + \frac{1}{12} m(m^2 + 3n^2 - 1).$$

Proof

Let $r_i = R_i + S_i$ where R_i is the rank of X_i among the X's and S_i the number of Y's smaller than X_i , then

$$M = 2 \sum_{i=1}^m S_i \left(R_i - \frac{m+1}{2} \right) - \sum_{i=1}^m S_i (n - S_i) + \frac{1}{12} m(m^2 + 3n^2 - 1),$$

$$M_{YXY} = \sum_{i=1}^m S_i (n - S_i),$$

$$M_{XYX} = \frac{1}{2} m n (m-1) - \text{the number of times an X appears between an X and a Y in the combined sequence of X's and Y's}$$

$$= \frac{1}{2} m n (m-1) - \sum_{i=1}^m \{ (R_i - 1)(n - S_i) + S_i (m - R_i) \}$$

$$= 2 \sum_{i=1}^m S_i \left(R_i - \frac{m+1}{2} \right)$$

which completes the proof.

The statistic M is thus a linear function of M_{XYX} and M_{YXY} ; however, for $m \neq n$ this linear function is unbalanced in the sense that M is not an estimator only of the dispersion parameters p_{YXY} , p_{XYX} but of a function in m and n of these parameters. This can easily be avoided by weighting the linear function correctly, namely.

$$M^* = (n-1) M_{XYX} - (m-1) M_{YXY},$$

for in this case

$$E(M^*) = \frac{1}{2} m n (m-1)(n-1)(p_{XYX} - p_{YXY}).$$

For symmetrical distributions only differing in location $p_{XYX} - p_{YXY} = 0$ and hence for this case the modified test based on M^* is no longer consistent against differences only in location. Clearly for $m = n$, M^* is a linear function of M and the two tests are equivalent.

In the case of $m = 2$, $n = 9$, considered above, the sequences which form the critical region of M^* are

X Y Y Y Y Y Y Y Y Y X
 X Y Y Y Y Y Y Y Y X Y
 Y X Y Y Y Y Y Y Y Y X
 X Y Y Y Y Y Y Y X Y Y
 Y Y X Y Y Y Y Y Y Y X

4. Properties of the test based on M^*

Since

$$\begin{aligned} & (n-1) \sum_{i_1 < i_2} \sum_j t(X_{i_1}, X_{i_2}; Y_j) - (m-1) \sum_{i_1} \sum_{j_1 < j_2} t(Y_{j_1}, Y_{j_2}; X_{i_1}) \\ & = \sum_{i_1 < i_2} \sum_{j_1 < j_2} \{ t(X_{i_1}, X_{i_2}; Y_{j_1}) + t(X_{i_1}, X_{i_2}; Y_{j_2}) - t(Y_{j_1}, Y_{j_2}; X_{i_1}) - t(Y_{j_1}, Y_{j_2}; X_{i_2}) \} \end{aligned}$$

and $t(a, b; c) + t(a, b; d) - t(c, d; a) - t(c, d; b) = \tau(a, b; c, d)$ where

$$\tau(a, b; c, d) = \begin{cases} 2 & \text{when } c \text{ and } d \text{ are between } a \text{ and } b \\ -2 & \text{when } a \text{ and } b \text{ are between } c \text{ and } d \\ 0 & \text{otherwise,} \end{cases}$$

M^* can also be written as

$$M^* = \sum_{i_1 < i_2} \sum_{j_1 < j_2} \tau(X_{i_1}, X_{i_2}; Y_{j_1}, Y_{j_2}).$$

It can also readily be shown that if $F \equiv G$

$$\sigma_{M^*}^2 = \frac{1}{45} mn(m-1)(n-1)(m+n+1)(mn-1).$$

The statistic M^* is proportional to a generalized $\bar{U}$ -statistic and from the theory of generalized $\bar{U}$ -statistics it follows, cf. Crouse (1960), that if $\lim_{m+n} \frac{m}{m+n}$ exists and is bounded away from one and zero then the limiting distribution of

$$\frac{M^* - E(M^*)}{\sigma_{M^*}}$$

is either normal, with finite variance, or degenerate at the origin. Specifically, for purposes of asymptotic relative efficiency, the following can be proved.

Theorem 3

If, as $(m+n)$ tends to infinity, $\lim_{m+n} \frac{m}{m+n}$ exists and is bounded away from one and zero,

$$G(x-\mu) = F\left\{(x-\mu)/\left(1 + \frac{c}{\sqrt{m+n}}\right)\right\}, \text{ where } \int x dF(x) = \int y dG(y) = \mu,$$

then the limiting distribution of $\frac{M^*}{\sigma_{M^*}}$ is normal with unit variance and mean

$$\lambda_{M^*} = \lim_{m+n \rightarrow \infty} \sqrt{\frac{45 mn}{m+n}} (p_{XYX} - p_{YXY}).$$

Under very general regularity conditions λ_{M^*} can be written as

$$\lambda_{M^*} = 2c \left(2 \int x F(x) f^2(x) dx - \int x f^2(x) dx \right) \lim_{m+n \rightarrow \infty} \left\{ \frac{45 mn}{(m+n)^2} \right\}^{1/2}$$

where $\frac{d}{dx} F(x) = f(x)$.

The corresponding value of the variance-ratio F-test can readily be shown to be

$$\lambda_F = \frac{2c}{(\beta_2 - 1)^{1/2}} \lim_{m+n \rightarrow \infty} \left\{ \frac{mn}{(m+n)^2} \right\}^{1/2}$$

where

$$\beta_2 = \frac{\int (x-\mu)^4 dF(x)}{(\int (x-\mu)^2 dF(x))^2},$$

and hence the asymptotic relative efficiency of the test based on M^* relative to the variance-ratio F-test is

$$\frac{\lambda_{M^*}^2}{\lambda_F^2} = 45 (\beta_2 - 1) \left\{ 2 \int x F(x) f^2(x) dx - \int x f^2(x) dx \right\}^2.$$

The latter is the same as the a.r.e. of the test based on M relative to the variance-ratio F-test (cf. Sukhatme (1957)).

Similarly it can be shown, cf. Crouse (1964), that the test obtained by applying M^* to the observations reduced by the sample medians is asymptotically distribution-free.

Finally the consistency properties of the test based on M^* can be summarized as follows:

Theorem 4

If $\lim_{m+n \rightarrow \infty} \frac{m}{m+n}$ exists and is bounded away from one and zero, and Z_α is the upper α -point of the standard normal distribution, then the test with critical region

$$\frac{M^*}{\sigma_{M^*}} > Z_\alpha$$

is consistent if and only if $p_{XYX} - p_{YXY} > 0$.

Proof

$$P \left\{ \frac{M^*}{\sigma_{M^*}} > Z_\alpha \right\} = P \left\{ \frac{M^* - E(M^*)}{\sigma_{M^*}} > Z_\alpha - \frac{E(M^*)}{\sigma_{M^*}} \right\}$$

and since

$$\frac{M^* - E(M^*)}{\sigma_{M^*}}$$

is asymptotically either normal with finite variance, or degenerate at zero, the result immediately follows from the fact that

$$\frac{E(M^*)}{\sigma_{M^*}} \rightarrow \text{infinity with } n \text{ if and only if } p_{XYX} - p_{YXY} > 0.$$

5. Critical values

The exact distribution of M^* was computed for $m = 2(1)10$, $n = m(1)10$ by enumerating all possible sequences of m X's and n Y's. The distribution was checked by comparing the computed mean and variance with the theoretical values. Critical values of M^* are presented in Table 1. Since M^* is a discrete variable, adjacent values M_1^* , M_2^* of M^* are presented together with $\alpha_1 = P\{M^* \leq M_1^*\}$, $\alpha_2 = P\{M^* \leq M_2^*\}$, where $\alpha_1 \leq \alpha \leq \alpha_2$. The nominal probabilities α were chosen as α , $1-\alpha = 0.005, 0.010, 0.025, 0.050, 0.100$. Thus one is able to perform one- and two-sided tests at the usual significance levels.

The normal approximation to the distribution of M^* was also computed, and the conclusion is essentially the same as that with respect to M which was reported by Laubscher et al (1968). For fairly large values of m and n the approximate exceedance probabilities may differ in the third decimal digit from the exact probabilities. An Edgeworth series approximation using the fourth moment would probably give a very good approximation.

6. Monte Carlo Study

In order to compare the modified test with Mood's test and the F-test, a small Monte Carlo study was performed. The purpose is to obtain an indication of the effect of difference in location on the power and significance level. Three distributions were chosen, namely uniform, normal and $\chi^2(2)$ -distributions. (None of the tests are strictly valid if a difference in location occurs if samples are drawn from the latter nonsymmetrical distribution.) The results are presented in Tables 2, 3 and 4. The tables give for various values of $\mu_X - \mu_Y$ and σ_Y^2 / σ_X^2 the number of times $H_0: \sigma_Y^2 = \sigma_X^2$ was rejected in favour of $H_a: \sigma_Y^2 > \sigma_X^2$ and $H_a: \sigma_Y^2 < \sigma_X^2$ out of 1000 samples, at a 10% (two-sided) significance level.

Table 2 (uniform distributions) yields the following information:

- (i) The F-test is conservative under H_0 , and, being by construction insensitive to differences in location, becomes more powerful than M and M^* when $\mu_X \neq \mu_Y$.
- (ii) Difference only in dispersion. Six sample power comparisons can be made between M^* and M in the case of the two-sided test. In all of these the M^* test performs a little better, the average ratio of the power of M relative to M^* being 0.96. In view of the fact that the critical region of M contains points of the sample space which do not neces-

sarily reflect differences in dispersion this result is not surprising.

- (iii) Difference only in location. The two-sided sample power of M^* (i.e. the error of the first kind) corresponding to sample sizes 4, 8; 4,12 and 4, 16 and a difference in location of 0.3 are

0.040, 0.043, 0.040

while the comparative figures for M are

0.040, 0.076, 0.141.

The similar figures for a difference in location of 0.6 are

M^* : 0.003, 0.003, 0.003

M : 0.004, 0.221, 0.483

These figures suggest M^* to be conservative against differences only in location and demonstrate the consistency of M against such alternatives, for $m \neq n$.

- (iv) Differences in location and dispersion. For a difference in dispersion the presence of a difference in location tends to reduce the power of the tests based on M^* and M . For a standard deviation ratio of 3 a difference in location of 0.3 reduces the power of M^* percentagewise by 15.7, 12.4, 6.4%

for sample sizes 4,8; 4,12 and 4,16 respectively. For M these figures are 20.9, 15.1, 14.5%

and for a difference in location of 0.6 the corresponding figures are

M^* : 36.0, 42.7, 45.0%

M : 46.0, 53.5, 61.7%

These figures suggest M^* to be somewhat more powerful than M when differences in location are present.

For Table 3 (normal distributions) the conclusions with respect to M and M^* are the same as for Table 2; the F-test is of course the most powerful test in this case.

Table 4 (chi-square distributions with two degrees of freedom) shows

- (i) the F-test is not valid, irrespective of differences in location.
- (ii) both M and M^* are sensitive to a difference in location, but M^* to a lesser extent than M .

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TABLE 1. CRITICAL VALUES OF MODIFIED MUOQ TEST
TOGETHER WITH EXACT CUMULATIVE PROBABILITIES

m	n	0.005	0.010	0.025	NOMINAL 0.050	PROBABILITY 0.100	0.900	0.950	0.975	0.990	0.995
2	3						2 0.9000	2 0.9000	2 0.9000	2 0.9000	2 0.9000
		-4 0.2000	-4 0.2000	-4 0.2000	-4 0.2000	-4 0.2000	6 1.0000	6 1.0000	6 1.0000	6 1.0000	6 1.0000
2	4						-8 0.0667	2 0.8000	6 0.9333	6 0.9333	6 0.9333
		-8 0.0667	-8 0.0667	-8 0.0667	-8 0.0667	-6 0.2000	6 0.9333	12 1.0000	12 1.0000	12 1.0000	12 1.0000
2	5						-12 0.0952	6 0.8571	6 0.8571	12 0.9524	12 0.9524
		-12 0.0952	-12 0.0952	-12 0.0952	-12 0.0952	-8 0.2381	12 0.9524	12 0.9524	20 1.0000	20 1.0000	20 1.0000
2	6						-18 0.0357	-18 0.0357	12 0.8929	20 0.9643	20 0.9643
		-18 0.0357	-18 0.0357	-18 0.0357	-16 0.1071	-16 0.1071	20 0.9643	20 0.9643	30 1.0000	30 1.0000	30 1.0000
2	7						-24 0.0556	18 0.8611	20 0.9167	30 0.9722	30 0.9722
		-24 0.0556	-24 0.0556	-24 0.0556	-24 0.0556	-20 0.1111	20 0.9167	30 0.9722	42 1.0000	42 1.0000	42 1.0000
2	8						-32 0.0222	-32 0.0222	30 0.8889	42 0.9333	42 0.9333
		-32 0.0222	-32 0.0222	-30 0.0667	-30 0.0667	-24 0.1556	30 0.9333	42 0.9778	42 0.9778	56 1.0000	56 1.0000
2	9						-40 0.0364	-32 0.0909	30 0.8909	42 0.9455	56 0.9818
		-40 0.0364	-40 0.0364	-40 0.0364	-36 0.0727	-30 0.1273	40 0.9091	56 0.9818	56 0.9818	72 1.0000	72 1.0000
2	10						-50 0.0152	-48 0.0455	-42 0.0758	38 0.8788	54 0.9242
		-50 0.0152	-50 0.0152	-48 0.0455	-42 0.0758	-40 0.1061	42 0.9091	56 0.9545	72 0.9848	90 1.0000	90 1.0000
2	11						-60 0.0256	-60 0.0641	-50 0.0874	52 0.9359	70 0.9615
		-60 0.0256	-60 0.0256	-60 0.0256	-56 0.0513	-48 0.1154	56 0.9231	72 0.9615	90 0.9872	110 1.0000	110 1.0000
2	12						-72 0.0110	-72 0.0110	-70 0.0530	-64 0.0549	-56 0.0566
		-72 0.0110	-72 0.0110	-70 0.0530	-64 0.0549	-54 0.1209	68 0.9121	90 0.9670	110 0.9850	132 1.0000	132 1.0000
2	13						-84 0.0150	-84 0.0150	-80 0.0381	-72 0.0381	-64 0.0381
		-84 0.0150	-84 0.0150	-80 0.0381	-72 0.0381	-64 0.0667	72 0.9048	108 0.9524	132 0.9905	156 0.9905	156 0.9905
2	14						-96 0.0083	-96 0.0083	-90 0.0250	-84 0.0417	-72 0.0583
		-96 0.0083	-96 0.0250	-90 0.0417	-84 0.0583	-72 0.1048	90 0.9167	130 0.9583	156 0.9917	182 0.9917	182 1.0000
2	15						-112 0.0147	-112 0.0147	-108 0.0294	-100 0.0441	-92 0.0515
		-112 0.0147	-112 0.0147	-108 0.0294	-100 0.0441	-92 0.0515	104 0.9118	132 0.9559	156 0.9779	182 0.9926	210 0.9926
2	16						-128 0.0065	-128 0.0065	-126 0.0196	-112 0.0458	-100 0.0850
		-128 0.0065	-126 0.0196	-120 0.0327	-110 0.0588	-96 0.1046	110 0.9020	156 0.9608	182 0.9804	210 0.9935	240 0.9935
2	17						-144 0.0117	-144 0.0117	-132 0.0251	-126 0.0326	-110 0.0409
		-144 0.0117	-144 0.0117	-132 0.0251	-126 0.0326	-106 0.1111	124 0.9006	178 0.9532	210 0.9825	240 0.9942	272 0.9942
2	18						-162 0.0053	-162 0.0053	-154 0.0158	-144 0.0474	-124 0.0947
		-162 0.0053	-154 0.0158	-140 0.0263	-140 0.0579	-120 0.1053	148 0.9105	206 0.9574	240 0.9842	272 0.9947	306 0.9947

TABLE 11(CONTINUED) CRITICAL VALUES OF MODIFIED MOOD TEST
TOGETHER WITH EXACT CUMULATIVE PROBABILITIES

m	n	0.005	0.010	0.025	NOMINAL 0.050	PROBABILITY 0.100	0.900	0.950	0.975	0.990	0.995
3	3						8 0.9000	8 0.9000	8 0.9000	8 0.9000	8 0.9000
		-12 0.1000	-12 0.1000	-12 0.1000	-12 0.1000	-12 0.1000	12 1.0000	12 1.0000	12 1.0000	12 1.0000	12 1.0000
3	4				-24 0.0286	-18 0.0857	16 0.8857	18 0.9429	18 0.9429	18 0.9429	18 0.9429
		-24 0.0286	-24 0.0286	-24 0.0286	-18 0.0857	-16 0.1429	18 0.9429	24 1.0000	24 1.0000	24 1.0000	24 1.0000
3	5				-36 0.0357	-28 0.0714	24 0.8929	28 0.9286	32 0.9643	32 0.9643	32 0.9643
		-36 0.0357	-36 0.0357	-36 0.0357	-28 0.0714	-24 0.1429	28 0.9286	32 0.9643	40 1.0000	40 1.0000	40 1.0000
3	6				-54 0.0119	-48 0.0357	40 0.8833	42 0.8929	44 0.9524	50 0.9762	50 0.9762
		-54 0.0119	-54 0.0119	-48 0.0357	-42 0.0595	-32 0.1071	40 0.9167	44 0.9524	50 0.9762	60 1.0000	60 1.0000
3	7				-72 0.0167	-60 0.0500	52 0.8833	60 0.9500	64 0.9667	72 0.9833	72 0.9833
		-72 0.0167	-72 0.0167	-60 0.0500	-56 0.0667	-44 0.1167	48 0.9167	64 0.9667	72 0.9833	84 1.0000	84 1.0000
3	8				-96 0.0061	-78 0.0424	62 0.0970	80 0.8727	84 0.9394	98 0.9879	98 0.9879
		-96 0.0061	-90 0.0182	-60 0.0303	-72 0.0545	-60 0.1091	60 0.9091	82 0.9515	88 0.9758	112 1.0000	112 1.0000
3	9				-120 0.0091	-108 0.0182	76 0.0909	96 0.9000	112 0.9455	128 0.9727	128 0.9909
		-120 0.0091	-108 0.0182	-104 0.0273	-84 0.0818	-76 0.1091	84 0.9273	104 0.9545	116 0.9818	144 0.9909	144 1.0000
3	10				-150 0.0035	-110 0.0245	92 0.0979	112 0.8916	138 0.9406	162 0.9860	162 0.9930
		-144 0.0105	-144 0.0105	-126 0.0315	-108 0.0559	-88 0.1049	96 0.9056	126 0.9545	144 0.9790	162 0.9930	180 1.0000
3	11				-180 0.0055	-156 0.0220	112 0.0879	132 0.8901	164 0.9341	184 0.9725	200 0.9945
		-180 0.0055	-168 0.0110	-152 0.0275	-132 0.0604	-108 0.1154	120 0.9066	144 0.9505	172 0.9780	220 0.9945	220 1.0000
3	12				-216 0.0022	-182 0.0242	136 0.0567	166 0.8923	198 0.9451	220 0.9868	224 0.9912
		-210 0.0066	-192 0.0154	-176 0.0286	-160 0.0527	-130 0.1011	140 0.9011	180 0.9582	200 0.9780	224 0.9912	242 0.9956
3	13				-252 0.0036	-208 0.0250	156 0.1000	196 0.8964	232 0.9464	264 0.9714	268 0.9893
		-240 0.0071	-228 0.0107	-200 0.0357	-180 0.0571	-144 0.1036	160 0.9036	204 0.9536	240 0.9821	268 0.9929	288 0.9964
3	14				-288 0.0044	-238 0.0235	172 0.0500	232 0.8956	268 0.9485	298 0.9735	316 0.9941
		-270 0.0074	-266 0.0103	-236 0.0265	-204 0.0588	-170 0.1044	184 0.9015	234 0.9515	274 0.9765	312 0.9912	338 0.9971
3	15				-324 0.0049	-244 0.0245	204 0.0680	260 0.8971	308 0.9436	336 0.9681	364 0.9926
		-308 0.0074	-300 0.0147	-272 0.0270	-240 0.0564	-200 0.1029	212 0.9044	264 0.9510	312 0.9779	348 0.9902	368 0.9951
3	16				-378 0.0031	-282 0.0248	228 0.0454	240 0.0980	304 0.8955	354 0.9732	420 0.9938
		-360 0.0052	-342 0.0114	-310 0.0268	-276 0.0516	-226 0.1001	244 0.9040	308 0.9515	360 0.9752	402 0.9917	424 0.9959
3	17				-420 0.0035	-316 0.0228	252 0.0439	272 0.1000	340 0.9491	396 0.9737	480 0.9947
		-400 0.0053	-392 0.0105	-352 0.0263	-312 0.0509	-248 0.1018	276 0.9018	348 0.9526	400 0.9754	448 0.9912	484 0.9965
4	4				-48 0.0143	-48 0.0143	36 0.0714	24 0.8714	30 0.9286	36 0.9286	36 0.9857
		-46 0.0143	-48 0.0143	-36 0.0714	-36 0.0714	-30 0.1286	30 0.9286	36 0.9857	36 0.9857	48 1.0000	48 1.0000

TABLE 1 (CONTINUED) CRITICAL VALUES OF MODIFIED WOOD TEST
TOGETHER WITH EXACT CUMULATIVE PROBABILITIES

m	n	0.005	0.010	0.025	NOMINAL 0.050	PROBABILITY 0.100	0.900	0.950	0.975	0.990	0.995
4	5			-72 0.0159	-56 0.0397	-44 0.0873	38 0.8889	52 0.9444	54 0.9603	60 0.9762	64 0.9921
		-72 0.0159	-72 0.0159	-60 0.0317	-54 0.0556	-42 0.1349	40 0.9048	54 0.9603	60 0.9762	64 0.9921	80 1.0000
4	6	-108 0.0048	-108 0.0048	-50 0.0238	-82 0.0429	-64 0.0857	58 0.8952	76 0.9429	80 0.9667	90 0.9857	90 0.9887
		-96 0.0143	-96 0.0143	-84 0.0333	-72 0.0571	-60 0.1048	60 0.9143	78 0.9524	86 0.9762	100 0.9952	100 0.9952
4	7		-144 0.0061	-126 0.0121	-108 0.0394	-84 0.0879	78 0.8939	102 0.9455	108 0.9636	120 0.9788	126 0.9909
		-144 0.0061	-126 0.0121	-120 0.0273	-102 0.0515	-78 0.1061	84 0.9152	108 0.9636	114 0.9758	126 0.9909	144 0.9970
4	8	-192 0.0020	-180 0.0061	-150 0.0222	-134 0.0444	-104 0.0990	102 0.8929	126 0.9475	148 0.9717	168 0.9899	174 0.9939
		-180 0.0061	-168 0.0101	-144 0.0283	-132 0.0523	-102 0.1030	104 0.9051	130 0.9515	154 0.9798	174 0.9939	196 0.9980
4	9	-240 0.0028	-216 0.0084	-186 0.0238	-162 0.0462	-128 0.0979	128 0.8965	162 0.9483	190 0.9748	210 0.9888	224 0.9930
		-216 0.0084	-210 0.0112	-180 0.0266	-160 0.0517	-126 0.1007	130 0.9049	164 0.9510	196 0.9776	216 0.9916	230 0.9958
4	10	-270 0.0050	-258 0.0090	-234 0.0220	-198 0.0450	-162 0.0959	156 0.8991	198 0.9491	228 0.9680	258 0.9880	270 0.9940
		-264 0.0070	-252 0.0130	-228 0.0260	-192 0.0569	-156 0.1015	162 0.9091	204 0.9590	234 0.9760	270 0.9940	288 0.9950
4	11	-330 0.0044	-308 0.0095	-272 0.0227	-234 0.0498	-188 0.0989	186 0.8960	238 0.9487	278 0.9744	316 0.9897	338 0.9941
		-324 0.0059	-294 0.0135	-270 0.0256	-232 0.0513	-186 0.1018	188 0.9004	240 0.9509	286 0.9758	326 0.9912	340 0.9956
4	12	-384 0.0049	-358 0.0099	-318 0.0236	-272 0.0495	-220 0.0984	228 0.8995	280 0.9500	330 0.9742	370 0.9879	402 0.9945
		-382 0.0060	-354 0.0121	-312 0.0269	-270 0.0538	-218 0.1005	230 0.9005	282 0.9511	332 0.9753	374 0.9901	414 0.9956
4	13	-450 0.0046	-426 0.0088	-372 0.0244	-324 0.0492	-258 0.0975	258 0.8975	330 0.9500	378 0.9748	426 0.9887	468 0.9950
		-444 0.0055	-414 0.0113	-366 0.0269	-318 0.0542	-252 0.1076	264 0.9021	336 0.9550	384 0.9752	444 0.9908	486 0.9958
4	14	-510 0.0049	-482 0.0098	-426 0.0245	-366 0.0500	-294 0.0984	302 0.8980	378 0.9484	442 0.9748	504 0.9899	546 0.9948
		-506 0.0056	-480 0.0105	-424 0.0258	-364 0.0520	-290 0.1003	304 0.9020	380 0.9503	448 0.9755	506 0.9905	552 0.9954
4	15	-594 0.0044	-550 0.0095	-482 0.0237	-420 0.0493	-334 0.0993	342 0.8999	430 0.9487	508 0.9750	580 0.9899	630 0.9948
		-582 0.0054	-548 0.0101	-480 0.0250	-414 0.0503	-332 0.1014	346 0.9014	432 0.9502	510 0.9760	584 0.9905	634 0.9954
4	16	-666 0.0045	-624 0.0097	-552 0.0248	-480 0.0487	-384 0.0958	384 0.8980	480 0.9461	564 0.9734	648 0.9897	708 0.9948
		-660 0.0058	-618 0.0105	-546 0.0256	-474 0.0504	-378 0.1011	390 0.9005	486 0.9507	570 0.9750	660 0.9903	720 0.9957
5	5		-120 0.0079	-104 0.0159	-56 0.0317	-72 0.0452	56 0.8889	72 0.9365	80 0.9683	96 0.9841	104 0.9921
		-120 0.0079	-104 0.0159	-96 0.0317	-80 0.0635	-64 0.1111	64 0.9048	80 0.9683	96 0.9841	104 0.9921	120 1.0000
5	6	-180 0.0022	-160 0.0065	-138 0.0238	-114 0.0476	-90 0.0931	88 0.8939	110 0.9459	128 0.9719	148 0.9870	150 0.9913
		-160 0.0065	-156 0.0106	-132 0.0260	-112 0.0519	-88 0.1104	90 0.9069	114 0.9545	130 0.9805	150 0.9913	160 0.9957
5	7	-240 0.0025	-208 0.0076	-184 0.0227	-156 0.0429	-124 0.0859	116 0.8914	148 0.9495	168 0.9747	204 0.9899	216 0.9949
		-216 0.0051	-204 0.0101	-180 0.0253	-152 0.0505	-120 0.1010	120 0.9066	152 0.9520	176 0.9758	212 0.9924	228 0.9975
5	8	-288 0.0039	-266 0.0085	-228 0.0249	-198 0.0490	-154 0.0963	154 0.8974	192 0.9495	224 0.9744	260 0.9883	276 0.9938
		-276 0.0054	-260 0.0101	-224 0.0264	-194 0.0521	-152 0.1010	156 0.9052	196 0.9510	230 0.9759	266 0.9915	288 0.9953

TABLE 1 (CONTINUED) CRITICAL VALUES OF MODIFIED MANN TEST
TOGETHER WITH EXACT CUMULATIVE PROBABILITIES

m	n	0.005	0.010	0.025	NOMINAL 0.050	PROBABILITY 0.100	0.900	0.950	0.975	0.990	0.995
5	9	-352 0.0050	-328 0.0100	-288 0.0220	-248 0.0460	-192 0.0999	184 0.8911	232 0.5431	272 0.5720	320 0.5690	344 0.5620
		-336 0.0070	-312 0.0140	-280 0.0280	-240 0.0529	-184 0.1109	192 0.9031	240 0.9530	280 0.9750	328 0.9910	352 0.9950
		-432 0.0043	-394 0.0100	-344 0.0250	-296 0.0480	-232 0.0976	232 0.8978	294 0.5487	342 0.5744	396 0.5890	440 0.5947
5	10	-420 0.0053	-350 0.0113	-340 0.0256	-254 0.0506	-230 0.1002	234 0.9004	296 0.9507	344 0.9757	402 0.9910	444 0.9953
		-512 0.0046	-468 0.0096	-416 0.0224	-348 0.0499	-276 0.0958	280 0.8988	348 0.5496	408 0.5744	476 0.5890	520 0.5945
		-504 0.0050	-464 0.0110	-408 0.0256	-344 0.0540	-272 0.1049	284 0.9025	352 0.9505	416 0.9753	480 0.9908	528 0.9954
5	12	-598 0.0048	-552 0.0099	-480 0.0242	-410 0.0496	-326 0.0992	326 0.8998	412 0.5496	490 0.5748	560 0.5898	616 0.5945
		-594 0.0052	-546 0.0102	-478 0.0255	-406 0.0503	-324 0.1015	328 0.9008	414 0.9502	492 0.9758	574 0.9901	618 0.9953
		-696 0.0047	-648 0.0098	-560 0.0250	-488 0.0465	-384 0.0971	376 0.8971	472 0.5461	560 0.5743	648 0.5897	712 0.5946
5	14	-688 0.0051	-640 0.0107	-552 0.0264	-480 0.0504	-376 0.1034	384 0.9024	480 0.9503	568 0.9753	656 0.9909	720 0.9956
		-798 0.0046	-738 0.0100	-640 0.0242	-550 0.0496	-438 0.0968	438 0.8993	554 0.5459	648 0.5745	750 0.5896	830 0.5945
		-792 0.0051	-736 0.0101	-638 0.0254	-548 0.0501	-436 0.1001	440 0.9010	558 0.9504	650 0.9752	752 0.9901	834 0.9953
5	15	-908 0.0046	-840 0.0098	-728 0.0249	-628 0.0491	-496 0.0995	500 0.8991	632 0.5496	740 0.5747	864 0.5898	952 0.5950
		-904 0.0050	-836 0.0103	-724 0.0254	-624 0.0515	-492 0.1019	504 0.9016	636 0.9514	744 0.9754	868 0.9901	956 0.9952
		-270 0.0011	-220 0.0097	-190 0.0238	-160 0.0465	-130 0.0963	110 0.8734	140 0.5307	170 0.5675	200 0.5848	220 0.5946
6	6	-240 0.0054	-210 0.0152	-180 0.0325	-150 0.0693	-120 0.1266	120 0.9037	150 0.9535	180 0.9762	210 0.9903	240 0.9989
		-312 0.0047	-290 0.0095	-246 0.0239	-212 0.0478	-166 0.0965	166 0.8980	208 0.5470	246 0.5745	288 0.5895	306 0.5948
		-306 0.0052	-280 0.0111	-244 0.0262	-210 0.0501	-162 0.1043	168 0.9050	210 0.9510	250 0.9773	294 0.9918	312 0.9959
6	8	-402 0.0043	-364 0.0100	-316 0.0240	-270 0.0456	-214 0.0976	212 0.8984	264 0.5474	310 0.5747	360 0.5887	396 0.5947
		-400 0.0053	-362 0.0107	-314 0.0253	-268 0.0503	-212 0.1002	214 0.9011	266 0.9500	312 0.9750	364 0.9900	402 0.9953
		-490 0.0050	-452 0.0098	-394 0.0244	-334 0.0498	-264 0.0953	264 0.8997	332 0.5459	394 0.5746	456 0.5898	498 0.5948
6	9	-486 0.0058	-450 0.0110	-352 0.0256	-330 0.0521	-262 0.1009	266 0.9033	334 0.9510	396 0.9762	458 0.9906	508 0.9952
		-598 0.0047	-550 0.0100	-476 0.0244	-406 0.0491	-320 0.0953	320 0.8977	406 0.5488	474 0.5748	556 0.5898	604 0.5945
		-594 0.0052	-548 0.0110	-474 0.0256	-404 0.0503	-318 0.1008	322 0.9010	408 0.9508	478 0.9750	558 0.9906	606 0.9951
6	11	-720 0.0046	-660 0.0099	-570 0.0245	-490 0.0489	-390 0.0968	370 0.8949	480 0.5459	560 0.5737	660 0.5895	720 0.5946
		-710 0.0053	-650 0.0110	-568 0.0260	-480 0.0520	-380 0.1033	380 0.9007	490 0.9541	570 0.9761	670 0.9912	730 0.9957
		-840 0.0048	-772 0.0100	-662 0.0245	-566 0.0499	-448 0.0955	450 0.8959	568 0.5491	668 0.5748	780 0.5898	856 0.5948
6	12	-836 0.0051	-768 0.0103	-660 0.0254	-564 0.0510	-446 0.1011	452 0.9016	570 0.9501	670 0.9751	782 0.9900	858 0.9953
		-978 0.0046	-896 0.0099	-772 0.0246	-658 0.0495	-522 0.0951	522 0.8953	662 0.5494	780 0.5748	908 0.5898	990 0.5950
		-974 0.0050	-854 0.0101	-770 0.0253	-656 0.0503	-520 0.1003	524 0.9006	664 0.9504	782 0.9751	912 0.9900	954 0.9952
6	14	-1116 0.0049	-1028 0.0098	-886 0.0246	-756 0.0499	-600 0.0951	598 0.8955	760 0.5496	896 0.5750	1042 0.5896	1138 0.5950
		-1114 0.0050	-1026 0.0101	-884 0.0251	-754 0.0504	-598 0.1002	600 0.9002	762 0.9502	898 0.9752	1044 0.9901	1144 0.9951

TABLE 1 (CONTINUED) CRITICAL VALUES OF MODIFIED MUO TEST
TOGETHER WITH EXACT CUMULATIVE PROBABILITIES

m	n	0.005	0.010	0.025	NOMINAL 0.050	PROBABILITY 0.100	0.900	0.950	0.975	0.990	0.995
7	7	-432 0.0029	-396 0.0082	-336 0.0233	-288 0.0466	-221 0.0950	204 0.8869	264 0.9452	312 0.9709	372 0.9889	396 0.9948
		-420 0.0052	-384 0.0111	-324 0.0291	-276 0.0548	-216 0.1121	216 0.9050	276 0.9534	324 0.9767	384 0.9918	420 0.9971
7	8	-524 0.0050	-466 0.0098	-418 0.0241	-356 0.0494	-280 0.0984	278 0.8995	352 0.9498	414 0.9739	478 0.9896	520 0.9946
		-522 0.0053	-464 0.0101	-416 0.0250	-354 0.0510	-278 0.1015	280 0.9016	354 0.9507	418 0.9751	482 0.9902	526 0.9952
7	9	-656 0.0049	-608 0.0093	-520 0.0248	-444 0.0491	-352 0.0969	344 0.8993	436 0.9493	512 0.9738	566 0.9888	660 0.9949
		-652 0.0051	-600 0.0103	-516 0.0259	-440 0.0507	-348 0.1004	348 0.9024	440 0.9514	516 0.9755	600 0.9902	664 0.9955
7	10	-798 0.0049	-728 0.0095	-626 0.0238	-540 0.0495	-426 0.0984	414 0.8986	528 0.9484	624 0.9747	726 0.9892	798 0.9949
		-792 0.0055	-732 0.0100	-630 0.0250	-534 0.0513	-420 0.1011	420 0.9001	534 0.9515	630 0.9759	732 0.9903	804 0.9953
7	11	-952 0.0048	-876 0.0099	-748 0.0246	-640 0.0486	-504 0.0991	500 0.8993	632 0.9491	746 0.9747	872 0.9896	956 0.9950
		-948 0.0050	-868 0.0101	-744 0.0254	-636 0.0502	-500 0.1011	504 0.9021	636 0.9504	752 0.9752	876 0.9901	960 0.9953
7	12	-1110 0.0050	-1022 0.0097	-878 0.0247	-746 0.0492	-550 0.1000	568 0.8997	746 0.9499	878 0.9749	1028 0.9900	1118 0.9950
		-1108 0.0050	-1020 0.0100	-876 0.0252	-744 0.0501	-548 0.1006	590 0.9007	748 0.9502	880 0.9752	1030 0.9903	1122 0.9952
7	13	-1296 0.0049	-1176 0.0093	-1022 0.0236	-876 0.0484	-656 0.0968	672 0.8981	852 0.9480	1008 0.9742	1176 0.9892	1296 0.9949
		-1264 0.0052	-1168 0.0101	-1020 0.0252	-864 0.0513	-684 0.1018	684 0.9025	864 0.9505	1020 0.9756	1168 0.9900	1308 0.9953
8	8	-686 0.0043	-644 0.0078	-546 0.0239	-462 0.0496	-364 0.0984	336 0.8908	434 0.9457	518 0.9740	616 0.9900	686 0.9942
		-672 0.0058	-630 0.0100	-532 0.0260	-448 0.0543	-350 0.1052	350 0.9016	448 0.9504	532 0.9761	630 0.9922	672 0.9957
8	9	-840 0.0049	-760 0.0100	-658 0.0250	-562 0.0497	-444 0.0992	440 0.8995	558 0.9498	654 0.9749	766 0.9900	832 0.9949
		-838 0.0051	-758 0.0103	-656 0.0255	-560 0.0502	-442 0.1003	442 0.9001	560 0.9504	656 0.9753	768 0.9902	834 0.9952
8	10	-1012 0.0050	-926 0.0099	-778 0.0245	-680 0.0494	-536 0.0995	532 0.8996	676 0.9498	798 0.9750	928 0.9900	1014 0.9949
		-1010 0.0051	-924 0.0104	-776 0.0251	-678 0.0500	-534 0.1002	534 0.9004	678 0.9503	800 0.9754	930 0.9901	1016 0.9950
8	11	-1208 0.0049	-1108 0.0099	-950 0.0247	-808 0.0497	-636 0.0996	634 0.8996	804 0.9499	948 0.9750	1106 0.9899	1210 0.9949
		-1206 0.0050	-1106 0.0100	-948 0.0251	-806 0.0502	-634 0.1004	636 0.9009	806 0.9501	950 0.9751	1108 0.9900	1212 0.9951
8	12	-1418 0.0049	-1296 0.0100	-1112 0.0249	-944 0.0499	-746 0.0996	742 0.8998	944 0.9500	1114 0.9749	1302 0.9900	1420 0.9949
		-1416 0.0050	-1296 0.0100	-1110 0.0253	-942 0.0504	-744 0.1001	744 0.9003	946 0.9504	1116 0.9752	1304 0.9901	1422 0.9951
9	9	-1056 0.0045	-976 0.0085	-822 0.0230	-704 0.0481	-560 0.0973	528 0.8975	672 0.9476	800 0.9742	944 0.9899	1024 0.9949
		-1040 0.0051	-960 0.0101	-816 0.0258	-688 0.0524	-544 0.1025	544 0.9027	688 0.9519	816 0.9770	960 0.9915	1040 0.9955
9	10	-1254 0.0050	-1150 0.0100	-986 0.0250	-836 0.0498	-656 0.0997	656 0.8997	836 0.9499	962 0.9748	1144 0.9900	1254 0.9950
		-1252 0.0051	-1148 0.0101	-984 0.0252	-834 0.0504	-656 0.1006	658 0.9004	838 0.9507	984 0.9752	1146 0.9901	1256 0.9950
9	11	-1456 0.0050	-1352 0.0098	-1172 0.0249	-956 0.0497	-784 0.0993	776 0.8990	988 0.9496	1168 0.9747	1364 0.9899	1456 0.9950
		-1442 0.0051	-1368 0.0100	-1168 0.0253	-992 0.0505	-780 0.1007	780 0.9003	992 0.9500	1172 0.9751	1368 0.9901	1500 0.9952
10	10	-1510 0.0050	-1404 0.0098	-1206 0.0241	-1026 0.0489	-816 0.0982	774 0.8966	960 0.9479	1170 0.9740	1368 0.9891	1464 0.9944
		-1512 0.0056	-1386 0.0105	-1188 0.0260	-1008 0.0521	-792 0.1034	792 0.9018	1008 0.9511	1188 0.9754	1386 0.9902	1512 0.9950

TABLE 2. NUMBER OF TIMES $H_0 : \sigma_y = \sigma_x$ WAS REJECTED IN FAVOUR OF
 $H_a : \sigma_y > \sigma_x$ (ALTERNATIVELY $H_a : \sigma_y < \sigma_x$) IN 1000 SAMPLES
 FROM UNIFORM DISTRIBUTIONS AT A 10% SIGNIFICANCE LEVEL.
 SAMPLE SIZES ARE m AND n , $\mu = \mu_x - \mu_y$, $\sigma = \sigma_y / \sigma_x$

m	n	TEST	$\mu=0$ $\sigma=1$	0 2	0 3	.3 1	.3 3	.6 1	.6 3
4	8	M*	53(49)	349(0)	572(0)	26(14)	482(0)	3(0)	366(0)
		M	58(45)	343(0)	550(0)	25(15)	435(0)	3(1)	297(0)
		F	23(18)	220(0)	544(0)	32(11)	500(0)	38(17)	512(0)
4	12	M*	39(41)	386(0)	651(0)	33(10)	570(0)	3(0)	373(0)
		M	37(45)	381(1)	609(0)	31(45)	517(0)	2(219)	279(2)
4	16	M*	50(62)	417(0)	687(0)	28(12)	643(0)	3(0)	368(0)
		M	41(63)	404(0)	640(0)	23(118)	547(0)	1(482)	244(3)

* * * * *

TABLE 3. NUMBER OF TIMES $H_0 : \sigma_y = \sigma_x$ WAS REJECTED IN FAVOUR OF
 $H_a : \sigma_y > \sigma_x$ (ALTERNATIVELY $H_a : \sigma_y < \sigma_x$) IN 1000 SAMPLES
 FROM NORMAL POPULATIONS AT A 10% SIGNIFICANCE LEVEL.
 SAMPLE SIZES ARE m AND n , $\mu = \mu_x - \mu_y$, $\sigma = \sigma_y / \sigma_x$

m	n	TEST	$\mu=0$ $\sigma=1$	0 2	0 3	1 1	1 3	2 1	2 3
4	8	M*	54(70)	257(1)	429(0)	28(36)	390(0)	2(3)	254(0)
		M	66(55)	249(0)	396(0)	20(40)	350(0)	3(8)	206(0)
		F	56(50)	291(1)	537(0)	49(56)	564(0)	46(46)	555(0)
4	12	M*	53(38)	260(0)	493(0)	18(40)	418(0)	3(8)	234(0)
		M	55(38)	263(1)	467(0)	18(71)	379(0)	1(249)	160(12)
4	16	M*	57(62)	286(1)	536(0)	12(31)	476(0)	0(4)	277(0)
		M	47(59)	275(0)	482(0)	6(120)	385(0)	0(510)	161(11)

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TABLE 4. NUMBER OF TIMES $H_0 : \sigma_y = \sigma_x$ WAS REJECTED (WHEN TRUE) IN
 FAVOUR OF $H_a : \sigma_y > \sigma_x$ (ALTERNATIVELY $H_a : \sigma_y < \sigma_x$) AT A 10%
 LEVEL IN 1000 SAMPLES FROM NONCENTRAL CHI-SQUARE DISTRIBUTIONS
 WITH 2 DEGREES OF FREEDOM. SAMPLE SIZES ARE m AND n , $\mu = \mu_x - \mu_y$

m	n	TEST	$\mu=0$	1	2
4	8	M*	50(48)	14(157)	1(144)
		M	60(42)	16(168)	1(176)
		F	132(106)	134(128)	143(122)
4	12	M*	56(50)	4(178)	1(165)
		M	57(56)	3(252)	1(443)
4	16	M*	45(37)	11(187)	3(182)
		M	45(32)	9(339)	3(568)